

Non-rational conformal field theory with $\mathfrak{sl}_3$ symmetry

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Joint work with Kazuya Kawasetsu and David Ridout

Overview

- 1 The Verlinde formula of rational CFT/VOAs
- 2 The standard module formalism: what and why
- 3 Weight modules for $\mathfrak{sl}_3$
- 4 A Verlinde formula for affine $\mathfrak{sl}_3$

The Verlinde formula of rational CFT/VOAs

Theorem [Huang, Moore-Seiberg, Verlinde]

The module category of a rational VOA is a modular tensor category.

Consequence: Verlinde formula

$$I_i \times I_j = \bigoplus_k N_{i,j}^k I_k,$$
$$N_{i,j}^k = \sum_{\ell} \frac{S_{i,\ell} S_{j,\ell} \overline{S_{k,\ell}}}{S_{\text{vac},\ell}},$$

where $S_{i,j}$ is the S -matrix in the basis of simple module characters.

The Verlinde formula and affine vertex algebras

- Kac-Wakimoto '88: Determined admissible levels and admissible highest weights for affine Lie algebras. Showed that modular group closes on characters of admissible modules.
- Kac-Gorelik '06: At admissible levels universal affine vertex algebras are reducible.
- Frenkel-Zhu '92: Simple quotient affine vertex algebras are rational precisely at non-negative integral levels
- Koh-Sorba '88: At non-integral admissible levels Verlinde formulae give negative fusion rules

Picking the right module category

For CFT the module category must satisfy:

- Closed under conjugation (contragredients)
- Closed under fusion
- Modules must have characters and characters should span a representation of the modular group

	rat $\mathcal{O}$	adm $\mathcal{O}$	adm KL	rel hw	spec flow & rel hw
conjugation	✓	×	✓	✓	✓
fusion	✓	×	✓	×	✓?
mod grp	✓	×	??	×	✓

The standard module formalism

Conjectural generalisation of the Verlinde formula.

Pioneered by Creutzig-Ridout. Further contributions from Canagasabay, Milas, Sato, Snadden, SW.

Procedure:

- 1 Identify *standard* modules.
- 2 Standards parametrised by measurable space (M, μ) .
- 3 Simple standards called typical. Reducible standards measure 0
- 4 The characters of standards form (topological) basis for all characters.
- 5 The characters of standards span representation of the modular group. In standard basis, the S-transformation is symmetric, unitary and squares to conjugation.
- 6 Plugging S-transformation into Verlinde formula gives non-negative integer fusion rules on Grothendieck group.

Standard module Verlinde formula

Let $R_\mu, \mu \in M$ be the standard modules, then the Verlinde product on the Grothendieck group is defined to be

$$[W_1] \times [W_2] = \int_M N_{1,2}^\mu [R_\mu] d\mu,$$

where

$$N_{1,2}^\mu = \int_M \frac{S_{W_1,\nu} S_{W_2,\nu} S_{\mu,\nu}^*}{S_{\text{vac},\nu}} d\nu.$$

Weight modules

Let $\mathfrak{g}$ be a simple finite dim Lie algebra with choice of Cartan subalgebra $\mathfrak{h}$ and root lattice Q .

Definition

A $\mathfrak{g}$ module M is called weight, if

$$M = \bigoplus_{\lambda \in \mathfrak{h}^*} M_{\lambda}, \quad \dim M_{\lambda} < \infty.$$

The set of λ such that $M_{\lambda} \neq 0$ forms the weight support.

Definition

A simple weight module M is called dense if its weight support fills out an entire coset in $\mathfrak{h}^*/Q$.

Weight modules

Theorem, Fernando '90

- Every simple weight module is either dense or simple quotient of induction from a simple dense module over a proper parabolic subalgebra.
- Only types A and C admit simple dense modules.

Theorem, Mathieu '00

Simple dense modules classified. Come in (coherent) families characterised by central character.

Theorem, Britten-Lemire-Futorny '95

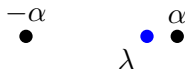
Simple dense $\mathfrak{sl}_3$ modules realisable by differential operators on spans of polynomials.

Highest weight $\mathfrak{sl}_2$ modules

$$\begin{array}{cc} -\alpha & \alpha \\ \bullet & \bullet \end{array}$$

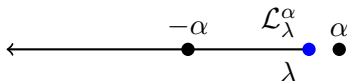
Different choices of positive roots all related by algebra automorphism twists.

Highest weight $\mathfrak{sl}_2$ modules



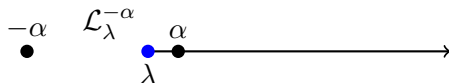
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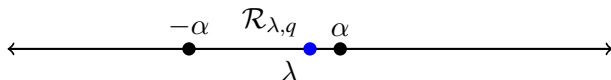
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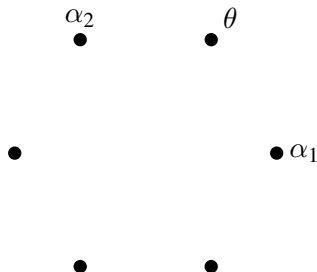
Isomorphism classes of dense modules are parametrised by weight coset and central character (quadratic Casimir eigenvalue q).

Dense $\mathfrak{sl}_2$ modules



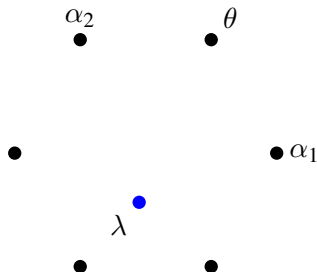
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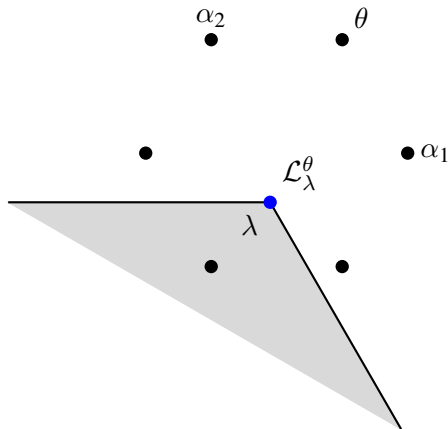
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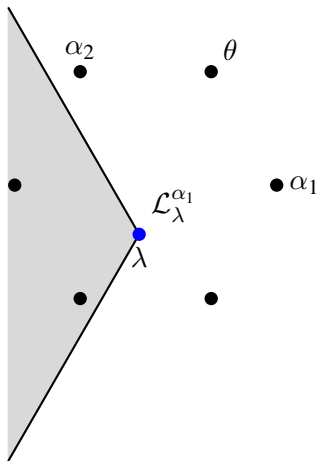
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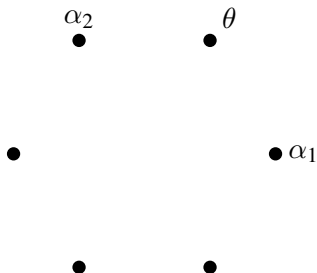
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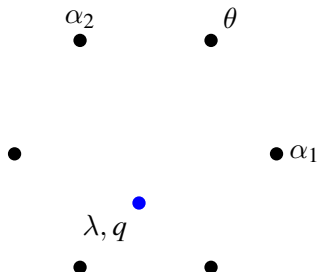
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Inductions from dense $\mathfrak{sl}_2$ modules



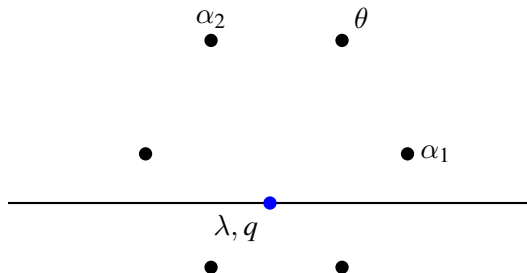
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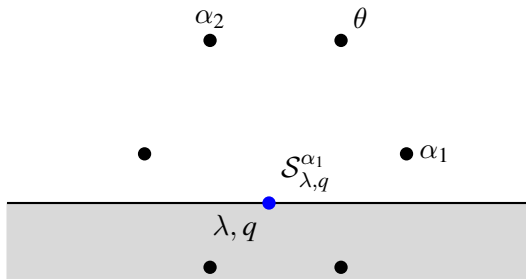
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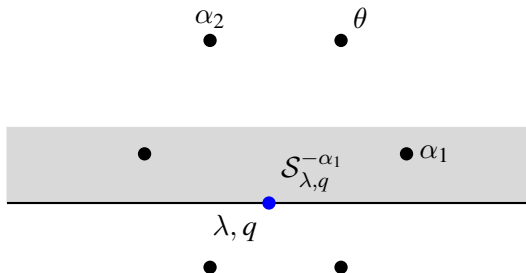
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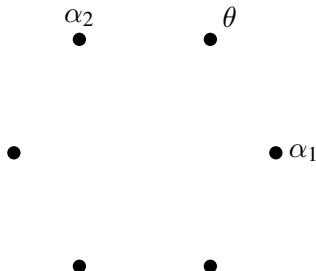
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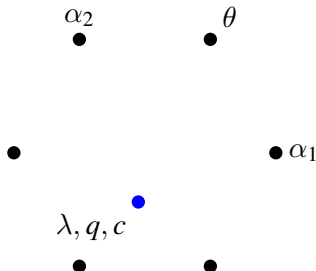
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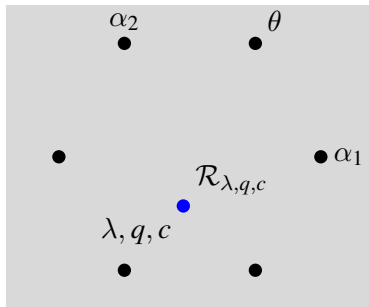
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Dense $\mathfrak{sl}_3$ modules



Isomorphism classes of dense modules are parametrised by weight coset and central character (quadratic, cubic Casimir eigenvalues q, c).

Affine $\mathfrak{sl}_3$ at level $k = -\frac{3}{2}$

- Smallest admissible level for $\mathfrak{sl}_3$ is $k = -\frac{3}{2}$
- Member of Deligne exceptional series
- Boundary admissible
- Trivial principal Hamiltonian reduction
- Minimal Hamiltonian reduction is rational BP algebra

Goals

- Classify simple $\mathbb{N}$ gradable weight modules from Zhu's algebra.
- Enlarge category to allow twists by algebra automorphisms.
- Compute characters and check that action of modular group closes.
- Compute fusion rules through standard module formalism version of Verlinde formula.

Affine $\mathfrak{sl}_3$ at level $k = -\frac{3}{2}$

Proposition

Universal affine vertex algebra $V^{-\frac{3}{2}}(\mathfrak{sl}_3)$ has singular vector χ at conformal weight 2 and $\mathfrak{sl}_3$ weight θ .

$$\chi = \left(3e_{-2}^{\theta} - 2h_{-1}^1 e_{-1}^{\theta} + 2h_{-1}^2 e_{-1}^{\theta} - 6e_{-1}^1 e_{-1}^2 \right) \Omega.$$

Proposition

- 1 $\text{Zhu}(V^{-\frac{3}{2}}(\mathfrak{sl}_3)) \cong U(\mathfrak{sl}_3)$
- 2 Image of singular vector in $\text{Zhu}(V^{-\frac{3}{2}}(\mathfrak{sl}_3))$
$$[\chi] = e^{\theta} (2h^2 - 2h^1 - 3) - 6e^2 e^1$$
- 3 $\text{Zhu}(L_{-\frac{3}{2}}(\mathfrak{sl}_3)) \cong U(\mathfrak{sl}_3) / \langle \chi \rangle$
- 4 Other generators of $\langle \chi \rangle$
$$[f_0^{\theta} \chi] = 6(f^1 e^1 - f^2 e^2) + h^1 (2h^1 + 3) - h^2 (2h^2 + 3)$$
$$[f_0^{\theta} f_0^{\theta} \chi] = -f^{\theta} (2h^1 - 2h^2 + 3) - 6f^1 f^2$$

Classifying highest weight $\text{Zhu}(L_{-\frac{3}{2}}(\mathfrak{sl}_3))$ modules

Proposition

- ① The highest weight module $\mathcal{L}_\lambda^\theta$ is annihilated by $\langle \chi \rangle$ if and only if

$$\lambda = (0, 0), \left(-\frac{3}{2}, 0\right), \left(0, -\frac{3}{2}\right), \left(-\frac{1}{2}, -\frac{1}{2}\right)$$

These are precisely the admissible weights for $k = -\frac{3}{2}$.

- ② Twisting by automorphisms gives (all) 13 highest weight $\text{Zhu}(L_{-\frac{3}{2}}(\mathfrak{sl}_3))$ modules.

Proof

Necessary conditions on highest weights given by acting with $[f_0^\theta \chi]$.
Sufficient conditions for all weights being realised given by acting with $[f_0^\theta f_0^\theta \chi]$.

Classifying inductions from dense $\mathfrak{sl}_2$ modules

Lemma

$$h^2(Q + \frac{3}{2}) \in \langle \chi \rangle$$

Proposition

- ① $\mathcal{S}_{\lambda,q}^{\alpha_1}$ is a $\text{Zhu}(L_{-\frac{3}{2}}(\mathfrak{sl}_3))$ module if and only if $q = -\frac{3}{2}$ and $(\lambda, \Lambda_2) = -\frac{1}{2}$. Simple if and only if $\lambda \notin -\frac{3}{2}\Lambda_1 + \mathbb{Z}\alpha_1, -\frac{1}{2}\theta + \mathbb{Z}\alpha_1$.
- ② Twisting by algebra automorphisms gives all five remaining 1-parameter families.

Classifying dense $\text{Zhu}(L_{-\frac{3}{2}}(\mathfrak{sl}_3))$ modules

Lemma

- 1 $C \in \langle \chi \rangle$
- 2 The centraliser of $\mathfrak{h}$ in $\text{Zhu}(L_{-\frac{3}{2}}(\mathfrak{sl}_3))$ is abelian, so all weight spaces of simple dense modules are 1-dimensional.

Proposition

- 1 $\mathcal{R}_{\lambda,q,c}$ is a $\text{Zhu}(L_{-\frac{3}{2}}(\mathfrak{sl}_3))$ module if and only if $q = -\frac{3}{2}$, $c = 0$.
Simple if and only if $(\lambda, \alpha^\perp) \neq \frac{1}{2}$ for all $\alpha \in \Delta$, where $\alpha^\perp = \alpha$ rotated by $\pi/2$ rescaled by $1/\sqrt{3}$.
- 2 All simple $\mathcal{L}$ and $\mathcal{S}$ type modules appear as composition factors of $\mathcal{R}$ modules (not true for general admissible levels).

Setting up the standard module formalism for $\mathfrak{sl}_3$

- The $\mathcal{R}_{\lambda, -\frac{3}{2}, 0} =: \mathcal{R}_\lambda$ and their spectral flows will form the standard modules.
- Characters can be computed from string function methods of Kawasetsu and Ridout.

Proposition

Let $\omega^i \in \mathfrak{h}$ satisfy $\alpha^i(\omega^j) = \delta_{i,j}$ and $\lambda = \lambda_1 \alpha^1 + \lambda_2 \alpha^2$.

$$\begin{aligned} \text{ch} [\mathcal{R}_\lambda] &= \text{Tr } y^K q^{L_0 - c/24} z_1^{\omega_1^1} z_2^{\omega_2^2} = \frac{y^k z_1^{\langle \lambda, \omega_1 \rangle} z_2^{\langle \lambda, \omega_2 \rangle}}{\eta(q)^4} \sum_{\beta \in Q} z_1^{\langle \beta, \omega_1 \rangle} z_2^{\langle \beta, \omega_2 \rangle} \\ &= \frac{y^{-\frac{3}{2}} z_1^{\lambda_1} z_2^{\lambda_2}}{\eta(q)^4} \sum_{m_1, m_2 \in \mathbb{Z}} z_1^{m_1} z_2^{m_2} \end{aligned}$$

Dense module characters

Recall the Dirac δ identity

$$\sum_{n \in \mathbb{Z}} e^{2\pi i \zeta n} = \sum_{m \in \mathbb{Z}} \delta(\zeta - m).$$

So

$$\sum_{\alpha \in Q} e^{2\pi i \langle \alpha, \zeta \rangle} = \sum_{\beta \in Q^*} \delta_{Q^*}(\zeta - \beta) =: \delta(\zeta = 0 \bmod Q^*), \quad \zeta \in \mathfrak{h},$$

where $\delta_{Q^*}(\zeta) = \prod_{i=1}^r \delta(\langle \alpha_i, \zeta \rangle)$.

Proposition

Let $y = e^{2\pi i \gamma}$, $z_j = e^{2\pi i \langle \alpha_j, \zeta \rangle}$, $q = e^{2\pi i \tau}$,

$$\text{ch} [\mathcal{R}_\lambda] (\gamma | \zeta | \tau) = \frac{e^{2\pi i \gamma k + 2\pi i (\zeta, \lambda)}}{\eta(q)^4} \delta(\zeta = 0 \bmod Q^*)$$

Spectral flow

Modular group does not close on span of $\text{ch} [\mathcal{R}_\lambda]$. Need to include spectral flow twists.

Proposition

$\text{Aut}_{\hat{\mathfrak{h}}}(\hat{\mathfrak{g}})$: Automorphisms of $\hat{\mathfrak{g}}$ which preserve $\hat{\mathfrak{h}}$.

$$\text{Aut}_{\hat{\mathfrak{h}}}(\hat{\mathfrak{g}}) \cong \text{Aut}(\mathfrak{g}) \ltimes Q^*$$

Elements in Q^* are called spectral flow.

$$\sigma^\beta(e_n^\alpha) = e_{n-\langle\alpha,\beta\rangle}^\alpha, \quad \sigma^\beta(h_n^\alpha) = h_n^\alpha - \langle\alpha,\beta\rangle\delta_{n,0}K, \quad \beta \in Q^*.$$

Twisting modules

Definition

Twisted module σM , where σ algebra automorphism.

$$x \cdot_{\sigma} m := \sigma^{-1}(x)m, \quad x \in \mathfrak{g}, \quad m \in M.$$

Corollary

$$\text{ch} \left[\sigma^{\beta} M \right] (\gamma | \zeta | \tau) = \text{ch} [M] (\gamma + \langle \beta, \zeta \rangle + \tau (||\beta||^2/2) | \zeta + \beta \tau | \tau)$$

Modular group action

Definition $\mathrm{SL}(2, \mathbb{Z}) = \langle S, T \mid S^2 = (ST)^3, S^4 = \mathbf{1} \rangle$

$$T : (\gamma|\zeta|\tau) \mapsto (\gamma|\zeta|\tau + 1),$$

$$S : (\gamma|\zeta|\tau) \mapsto \left(\gamma - \frac{||\zeta||^2}{2\tau} \mid \frac{\zeta}{\tau} \mid -\frac{1}{\tau} \right).$$

Proposition

$$T \left\{ \mathrm{ch} \left[\sigma^\beta \mathcal{R}_\lambda \right] \right\} = e^{2\pi i(-1/6 + (\lambda, \beta) - 3/2||\beta||^2)} \mathrm{ch} \left[\sigma^\beta \mathcal{R}_\lambda \right]$$

$$S \left\{ \mathrm{ch} \left[\sigma^\beta \mathcal{R}_\lambda \right] \right\} = \sum_{\phi \in Q^*} \int_D S_{\beta, \lambda; \phi, \lambda'} \mathrm{ch} \left[\sigma^\phi \mathcal{R}_{\lambda'} \right] d\lambda',$$

where $D = \{ \lambda = \lambda_1 \alpha_1 + \lambda_2 \alpha_2 \mid -1/2 \leq \lambda_i \leq 1/2 \}$

$$S_{\beta, \lambda; \phi, \lambda'} = \frac{|\tau|^2}{(-i\tau)^2} e^{2\pi i(-(\beta, \lambda') - (\phi, \lambda) + 3/2(\beta, \phi))}.$$

Verlinde algebra on Grothendieck group

$$\begin{aligned} [\sigma^\beta \mathcal{R}_\lambda] \times [\sigma^\phi \mathcal{R}_{\lambda'}] = & 2[\sigma^{\beta+\phi} \mathcal{R}_{\lambda+\lambda'}] \\ & + [\sigma^{\beta+\phi+\omega_1} \mathcal{R}_{\lambda+\lambda'-3\omega_1/2}] + [\sigma^{\beta+\phi-\omega_1} \mathcal{R}_{\lambda+\lambda'-3\omega_1/2}] \\ & + [\sigma^{\beta+\phi+\omega_2} \mathcal{R}_{\lambda+\lambda'-3\omega_2/2}] + [\sigma^{\beta+\phi-\omega_2} \mathcal{R}_{\lambda+\lambda'-3\omega_2/2}] \\ & + [\sigma^{\beta+\phi+\varepsilon} \mathcal{R}_{\lambda+\lambda'+\theta/2}] + [\sigma^{\beta+\phi-\varepsilon} \mathcal{R}_{\lambda+\lambda'+\theta/2}], \end{aligned}$$

where $\varepsilon = (\alpha_1 - \alpha_2)/3$.

Similar formulae can be computed for all other modules. All fusion coefficients are non-negative integers.

The End

Thank you for coming!